Rates

1 Tricia reads 20 pages of her book in 2 hours. Find her reading rate of pages per hour.

2 A car travels $320\ \text{km}$ in 4 hours.

a Complete the following table of values:

Time taken (hrs)	4	2	1
Distance travelled (km)	320		

b Find the speed of the car in kilometres per hour.

3 A tap fills up a 240 litre tub in 4 hours.

a State the compound unit for the rate of water flow.

b Find the rate of water flow of the tap.

4 A toy manufacturer makes 720 toys in 9 hours.

a State the compound unit for the rate of toy manufacturing.

b Find the rate at which the toy manufacturer works.

5 Justin earns $\$420$ in 6 hours. Find Justin's hourly rate.

6 Tina burns 640 calories by jogging 8 kilometres. Find the rate at which Tina burns calories per kilometre.

7 Luke writes 900 words in 20 minutes. Find Luke's writing speed in words per minute.

8 A 3-episode TV series had a total run time of 69 minutes. Find the run time for a single episode.

9 An animator uses 12 frames for every 0.5 seconds of an animation. Find the frame rate of the animation in frames per second (fps).

10 Avril is running from John in a game of tag. Avril runs 30 metres every 5 seconds and John runs 21 metres every 3 seconds.

a Find Avril's speed in metres per second.

b Find John's speed in metres per second.

c Who is faster?

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Applications

11 Iain earns $\$22$ per hour working as a performer and works for 20 hours a week. Find Iain's weekly earnings.

12 Caitlin plays cricket over the summer holidays. Caitlin scores at a rate of 24 runs per game and there are 30 games in the season. How many runs did Caitlin score in total?

13 A bus travels at a speed of $80\ \text{km/hr}$. How far does the bus travel in 5 hours?

14 It requires $8\ \text{mL}$ of polyethylene to create 20 plastic bags.

- **a** Find the rate of polyethylene used per plastic bag.
- **b** How much polyethylene would be required to create 5 plastic bags?

15 A worm takes 15 seconds to travel $20\ \text{cm}$.

- **a** How long does it take for the worm to travel 1 metre?
- **b** Find the rate of the worm's time spent per metre distance traveled.

16 A tap is dripping at a rate of $1000\ \text{mL/hr}$. A $5\ \text{L}$ bucket is placed underneath it to catch the water.

- **a** How many millilitres of water can the bucket hold?
- **b** How many hours will it take to fill the bucket?

17 Lisa is trying to save money so she deposits $\$3.20$ into her bank account for every 8 coffees she purchases.

How much does Lisa deposit for each cup of coffee?

18 Patricia eats 7.2 litres of ice cream in 6 minutes in an ice-cream eating contest. Patricia wants to find her rate of ice-cream consumption in millilitres per second.

- **a** How many millilitres of ice-cream did Patricia consume?
- **b** How many seconds did it take for Patricia to consume the ice-cream?
- **c** Find her rate of consumption of ice-cream in mL/s.

19 A cyclist travels 70 metres per 10 seconds. Find the speed of the cyclist in metres per minute.

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20 Tina counts 26 heart beats over a 20 second interval. Find Tina's heart rate in beats per minute (bpm).

21 Adam really likes apples and eats them at a rate of 4 per day.

a How many apples does Adam eat in one week?

b If Adam buys 44 apples how many days will this last him?

22 The height of a pool increases by $5\ \text{cm}$ for every 10 litres of water added. After adding some water the height of the pool increased by $0.45\ \text{m}$.

How much water was added to the pool?

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